# Software Requirements Specification

for

# HHH – The Ride App

Version 1.0


Prepared by

**Jon Scales, Tim Haxton, Cooper Wolf**

**Kings of Code**


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Revision History

| Name | Date | Reason For Changes | Version |
|---|---|---|---|
| TH, CW, JS | 3/18/25 | Initial Document Submission | 1.0 |
| | | | |

# 1. Introduction

## 1.1 Purpose

The product whose software requirements are specified in this document is HHH – The Ride App, v1.0. The scope of the project is to allow organizers for Hotter'N Hell (HHH) in Wichita Falls, TX to register riders on the day of an event and manage the race and rider databases, to allow riders to request Support and Gear (SAG) assistance, and to provide users of the application with information regarding the events held on HHH weekend and other attractions within the Wichita Falls local area.

## 1.2 Document Conventions

Priority Assignment

Application features are given priority ratings of 1-10, where 1 represents the HIGHEST priority and 10 represents the LOWEST priority.

## 1.3 Intended Audience and Reading Suggestions

This document is intended to be read by the developers and documentation writers of the HHH- The Ride App project and the primary stakeholder, Dr. Eduardo Colmenares. The rest of this SRS contains an overall description, external interface requirements, system features, and nonfunctional requirements for HHH – The Ride App. It is imperative that this document be read cover to cover, in the order that it is written, for clarity and understanding.

## 1.4 Product Scope

HHH – The Ride App is intended to be a mobile application for iOS, Android, and Windows. It is race event management software, primarily designed for riders and organizers of the HHH cycling event. It will provide an interface for riders to register for a HHH event on the day of the event, request SAG assistance, and view a schedule of events and attractions occurring during the weekend of the race, within the Wichita Falls local area. It will also allow organizers to manage rider databases.

## 1.5 References

Feasibility_Study_HHH.docx, The Kings of Code, ver 1.0, March 4, 2025, https://github.com/jonscales/5153-SWE-Project/blob/main/hhh-app/Feasibility_Study_HHH.docx

# 2. Overall Description

## 2.1 Product Perspective

This product is designed to be a replacement for the current on-site HHH registration system. It will be designed to free volunteers and accelerate on-site registration by allowing users to register for HHH events using a mobile device on the day of the event. It will also allow riders to request SAG assistance during an event, which is intended to provide faster response for event participants. Additionally, the app will provide event schedules and local area information to app users who may be unfamiliar with the local area or are curious about things to do in Wichita Falls during HHH weekend. The app is intended for use by all visitors, participants and HHH personnel during the week/weekend of the HHH event.

## 2.2 Product Functions

1. On-site event registration
2. Participant database
3. BikeReg database merge
4. Schedule of events
5. Calendar alerts
6. SAG request
7. Route maps
8. Tradeshow vendors list
9. Local restaurant and lodgings list

## 2.3 User Classes and Characteristics

1. HHH Administrator/Volunteer
    - Will have access to all app functions
2. HHH Participant/Rider
    - Will have access to all app functions except database functions
3. HHH Spectator/App User
    - Will be able to register for events and view event info and weekend schedules
    - Will not have access to database functions or SAG request

## 2.4 Operating Environment

The app will operate within the following operating systems and devices:

- iOS 18+ on iPhone X or newer devices
- Android 14+ on Android mobile devices
- Windows 11 on Windows PCs

## 2.5 Design and Implementation Constraints

Issues that could limit the options available to developers are as follows:

1. Language Requirements:
    - Frontend development restricted to JavaScript, Python, DART, or Django
    - Backend development restricted to C++, Python, or Java
    - Database restricted to SQL
2. Funds Limitations:
    - As college students, we are limited in financial support to our own personal capabilities. All required databases and/or IDEs must be low- or no-cost. This will lead to an associated limitation on available space.

## 2.6 User Documentation

The intuitiveness of The Ride App user interface makes user documentation needs extremely minimal. The following paragraph is what will appear in the Google Play store and the Apple App store to describe the installation, use and functionality of The Ride App.

The Ride App is for use by all participants, spectators, and volunteers of the HHH. Downloading the app allows users to register on-site for events, view HHH weekend schedules, tradeshow vendors, local restaurants, route maps, and much more. The Ride App will require access to your mobile device's clock, location, phone, calendar, and map application.

## 2.7 Assumptions and Dependencies

Use of The Ride App makes the assumption that the app will be granted the necessary permissions of access to the mobile device's clock, location, phone, calendar and map applications. The Ride App is dependent on access to the device's location in order utilize the SAG request and local information features. The Ride App is dependent on access to the device's phone to utilize the SAG call feature. The Ride App is dependent on access to the device's map application (google maps/apple maps) to utilize the local information feature. The Ride App is dependent on access to the device's calendar application to utilize the event schedule reminder feature. The Ride App is dependent on access to the clock and calendar to utilize the event registration feature ensuring only available events are listed for registration.

# 3. External Interface Requirements

## 3.1 User Interfaces

### 3.1.1 Introduction

The application provides a user-friendly interface designed to be accessible for users of all ages. The UI follows a structured and intuitive layout with clearly labeled buttons, easy-to-follow instructions, and informative content. The design incorporates a visually appealing theme inspired by the HHH event, using a consistent color scheme of yellow, red, and orange. The application will be available for both iOS and Android devices, ensuring cross-platform compatibility.

### 3.1.2 Description of UI Components

Main Screen – The primary screens within the application include:

1. Welcome/Login Screen – Allows users to enter their first name, last name, and email to proceed.
2. Main Menu – Provides access to key functionalities, including:
   a. Info & Events
   b. On-Site Registration
   c. Volunteer Login (for administrators)
3. Registration Page – Displays event options for users to register.

4. Events & Info Page – Offers navigation to event schedules, trade show vendors, and maps.
5. Maps Page – Allows users to view different cycling route, including:
    a. All endurance routes
    b. All gravel routes
    c. Mountain bike route
    d. 10K route
6. Schedules Page – Provides buttons for viewing:
    a. Ride start times
    b. Trade show hours

Navigation Flow

1. Users enter their first name, last name, and email on the Welcome/Login screen.
2. Clicking submit navigates users to the Main Menu.
3. The Main Menu presents the following options:
    a. Info & Events – allows users to view HHH event details
    b. On-Site Registration – allows riders to register for event
    c. Volunteer Login – allows admin to view database
4. The Events & Info page contains buttons leading to:
    a. Event schedule
    b. Trade Show Vendors
    c. Maps – allows users to view maps of all events
5. The Schedule Page includes buttons for ride start times and trade show hours.

Forms and Inputs

1. All users must enter their first name, last name, and email on the Welcome/Login Screen to access the app.
2. Administrators and volunteers provide additional credentials through the Volunteer Login section.
3. Potential riders will enter information specific to the race they are registering for.

### 3.1.3 Usability and Accessibility Considerations

1. Cross-Platform Compatibility: Designed for iOS, Android and Windows.
2. Clear Button Labels: Ensuring ease of navigation for all users.
3. Readable Font and High Contrast: Enhancing readability for different age groups.

### 3.1.4 Constraints and Assumptions

1. The application must support devices with a minimum screen resolution of 720p.
2. The UI will include touch support for mobile devices.

## 3.2 Hardware Interfaces

### 3.2.1 Introduction

The app is designed to operate on modern mobile and desktop devices, including smartphones, tablets, and computers running supported operating systems. It requires internet connectivity for full functionality.

### 3.2.2 Supported Devices

- Smartphones (iOS 18+, Android 14+)
- Tablets (iOS 18+, Android 14+)
- Desktop/Laptops (Windows 11)

### 3.2.3 Minimum Hardware Requirements

- Processors – 1.8 GHz dual-core
- RAM – 3GB
- Storage – 100MB of free space for installation
- Screen Resolution - 720p
- Internet Connectivity – Wi-Fi or cellular data

### 3.2.4 Minimum Hardware Requirements

- The app assumes users will have stable internet connection for real-time updates.

## 3.3 Software Interfaces

### 3.3.1 Frontend (Flutter/FlutterFlow)

The frontend is developed using Flutter and FlutterFlow. It is responsible for providing the user interface, managing user interactions, and displaying data retrieved from the backend.

Communications:

- Flutter communicates with the backend using RESTful API endpoints over HTTP/HTTPS.
- Data is exchanged in JSON format.
- Flutter will make requests like
  - GET – fetches data (event details, user profiles)
  - POST – submit forms (event registration)
  - PUT/DELETE – updates or removes resources

### 3.3.2 Backend (Python with FastAPI)

FastAPI is used as the backend framework to implement the business logic, manage authentication, process data, and serve as the intermediary between the frontend and the database.

API Endpoints:

- The backend exposes a series of RESTful endpoints that are consumed by the Flutter frontend.
- Endpoints are documented and available via FastAPI's built-in Swagger UI

Authentication & Security:

- The backend employs secure authentication mechanisms to ensure that only authorized users access specific endpoints.

Data Processing:

- Incoming requests are processed, and the appropriate business logic is executed.
- Responses are generated in JSON format for consumption by the frontend.
- FastAPI handles CRUD operations with the MySQL database, ensuring data consistency and integrity.

### 3.3.3 Database (MySQL)

MySQL serves as the relational database system for persistent data storage. It manages critical application data such as rider and admin information as well as event registrations.

Integration with Backend:

- FastAPI connects to MySQL using an ORM (Object-Relational Mapping)
- All database transactions are mediated through FastAPI, which ensures that data is validated and sanitized before any operations are performed.

Data Structure and Integrity:

- The database schema is designed with appropriate constraints including primary keys, foreign keys, and unit indexes.
- Data retrieved from MySQL is formatted as JSON by the backend before being sent to the Flutter frontend.
- SQL queries are used for efficient data manipulation and retrieval, ensuring consistency and performance.

## 3.4 Communications Interfaces

### 3.4.1 Introduction

This section defines the communication protocols and data exchange mechanisms used by the app. The app requires internet connectivity for real-time data retrieval, event updates, and user interactions. All data exchanges are handed securely between the Flutter frontend, FastAPI backend and MySQL database.

### 3.4.2 Network Protocols

The application relies on the following network protocols for communication:

- HyperText Transfer Protocol Secure (HTTPS) - Used for secure API requests between the frontend and backend.
- TCP/IP - Ensures reliable data transmission between the backend and MySQL database.

### 3.4.3 Data Exchange & API Communication

Flutter communicates with the FastAPI through:

- RESTful API (via FastAPI) - The backend provides REST endpoints for user authentication, event registration and data retrieval.
- JSON Format – Used for structured data exchange between frontend and backend.

FastAPI interacts with MySQL using:

- SQL Queries – Data is retrieved, inserted, or updated via SQL queries executed by FastAPI.

### 3.4.4 Constraints and Assumptions

- A stable internet connection is required for real-time updates and event registration.
- The API must be available and responsive for proper app functionality.
- MySQL must maintain high availability to prevent data loss or service disruption.

# 4. System Features

## 4.1 Streamlined on-site participant registration for any HHH event.

### 4.1.1 Description and Priority

***The priority of this feature is 1***

App users can download the app and easily register for events on-site beginning on Thursday prior to the weekend events. Events requiring registration begin on Friday morning with off-road bicycle races and run through Sunday afternoon with professional criterium races.

### 4.1.2 Stimulus/Response Sequences

Potential application users shall be able to download the mobile app from links provided on the HHH website, QR codes posted throughout the event venue, the Apple app store, or google play. Upon opening the app, users will confirm permissions granting app access to make phone calls, calendar, and location. Users will enter their name and email address. The app will automatically access the phone number. These items of user information will be available to populate registration form fields if the user registers for any event.

### 4.1.3 Functional Requirements

The following are functional requirements to implement feature 1 of our application.

REQ-1-1: The Ride App shall collect the following information as a form from all registering, ride participants: first name, last name (collected via initial landing page fields), age, email address (collected via initial landing page field), phone number, city or country, event being registered for, emergency contact full name, emergency contact phone number.

REQ-1-2: The Ride App shall also collect gender (male or female) and category of racer (cat 1, cat 2, novice) within forms for participants registering for racing events (off-road races, trail runs).

REQ-1-3: The Ride App shall only make available to registrants those events which are upcoming and for which registration may still occur. Access to the device's calendar and clock allows The Ride App to turn off expired events.

## 4.2 Database of registered participants

### 4.2.1 Description and Priority

***The priority of this feature is 1***

The HHH event requires a list of all participants. A majority of participants register prior to the week of the events through the 3rd party provider BikeReg.com. These participants are termed "pre-registered". Some participants will register shortly before the actual start of the event, these participants are termed "on-site registrants". The on-site registrants will utilize the HHH-The Ride App rather than the BikeReg.com platform. The on-site participant information shall be stored in a database.

### 4.2.2 Stimulus/Response Sequences

Only HHH volunteers and administrators shall have access to the database portion of the application. When participants submit a registration form for an event, the registration form data will be uploaded into a SQL database. This process will be handled automatically through the app software.

### 4.2.3 Functional Requirements

REQ-2-1: The application shall upload to the database the information collected via the registration form fields of the registration pages of the app. This data will include ride participant's first & last name, age, email address, phone number, city, zip code, or country, event being registered for, emergency contact first & last name, emergency contact phone number, gender*, category of racer*, *(these will be null values for non-race participants).

REQ-2-2: The app shall use initial app setup information and event selection to autofill the name, email, and event fields of the registration forms thereby streamlining the registration process.

## 4.3 Ability to initiate The Ride App database from pre-existing BikeReg database.

### 4.3.1 Description and Priority

***The priority of this feature is 1***

The database of registered participants will be downloaded from the BikeReg Event Manager control page as a CSV file. The registration of additional participants through the BikeReg site will be terminated at that point. The CSV will be uploaded into The Ride App's SQL database as an initial dataset and our app will continue to add applicable registrant information. The goal is to have a single database of HHH participants with information suitable to allow association of riders with their bib numbers in the case of emergency situations,

### 4.3.2 Stimulus/Response Sequences

This aspect of The Ride App will be a backend process initiated by the HHH administration users.

### 4.3.3 Functional Requirements

REQ-3-1: The Ride App shall provide administrative/volunteer users with a login menu choice allowing them to enter a username and password for authentication.

REQ-3-2: The Ride App shall restrict database management functions to admin/volunteer users via password authentication.

REQ-3-2: The desktop/tablet version of The Ride App shall provide a way for an administrator to upload a CSV containing all preregistered participants.

## 4.4 Schedule of individual events for the entire weekend

### 4.4.1 Description and Priority

***The priority of this feature is 2***

There are multiple races, rides and running events throughout the course of the weekend. There is also a tradeshow for vendors selling a variety of biking/running goods. It will be nice for the app to allow the user to view a consolidated list of these events along a timeline.

### 4.4.2 Stimulus/Response Sequences

The app will have a schedule of events button the user can tap to bring up a timeline or calendar view of scheduled events showing the event's starting time (duration if applicable) and location.

### 4.4.3 Functional Requirements

REQ-4-1: The application shall have a menu button to open text list of scheduled events.

## 4.5 Ability for a user to create personalized event reminders in calendar.

### 4.5.1 Description and Priority

***The priority of this feature is 6***

This is a low priority feature because the design team is unsure if this feature may be technically impossible to implement due to unforeseen limitations that may exist in having The Ride App communicate with other apps and/or functions on their mobile device such as the endogenous calendar app present on their phone.

### 4.5.2 Stimulus/Response Sequences

This feature would be initiated after opening the scheduled events list and tapping any particular event. The app would open the calendar app on the user's device and initiate the addition of a new appointment/event that would then be imported to their device's calendar schedule, and which would allow them to set a reminder via their scheduling app.

### 4.5.3 Functional Requirements

REQ-5-1: The Ride App shall allow users to click a particular event to initiate generation of a reminder on their mobile device.

REQ-5-2: The Ride App shall access and open the device's existing calendar/scheduler app to allow a way for the user to add the event to the calendar app on their mobile device.

## 4.6 Ability of participant to contact SAG volunteers.

### 4.6.1 Description and Priority

***The priority of this feature is 2***

Riders which experience mechanical problems or for other reasons can or wish not to complete their ride are assisted by HHH volunteers through the event SAG operations. SAG personnel will pick up riders along the ride routes or attempt repairs to rider's equipment. Some riders do experience non-life-threatening medical issues and may require SAG assistance to return to the ride start or medical facilities. In any of these instances, the ability of a rider to rapidly communicate to the SAG dispatch center volunteers is a desired feature of this application.

### 4.6.2 Stimulus/Response Sequences

Should a rider require SAG assistance, they can quickly do so by tapping the CALL SAG button from The Ride App. When installed, the app has permission to access a mobile device's phone functions. This will allow the SAG dispatch center phone number to be directly entered for dialing when the CALL SAG button is pressed. This alleviates the need for a rider to look up what number to call. The app will instruct riders to dial 911 rather than SAG for any extreme medical emergency situations.

### 4.6.3 Functional Requirements

REQ-6-1: The Ride App shall have a screen with a button to direct dial the SAG dispatch center along with instructions to dial 911 if there is a medical emergency.

REQ-6-2: The Ride App shall be able to access the mobile device phone functionality to auto fill the SAG dispatch phone number to be dialed.

## 4.7 Access to route maps for all rides

### 4.7.1 Description and Priority

***The priority of this feature is 2***

Riders will want to see and study the ride routes. Many riders track their rides via apps such as Strava, but such apps do not show a preset route rather, only a track of where the rider has been. In addition to just maps of the ride routes, The Ride App will have maps of the event venue parking, start line staging positions, start line access, and the off-road event trails.

### 4.7.2 Stimulus/Response Sequences

The app will have a maps dropdown menu with buttons to link to enlargeable map images which can be panned to see the entire route in detail even on a mobile device. Image enlargement and panning will be done with standard pinch-to-zoom touchscreen interaction.

### 4.7.3 Functional Requirements

REQ-7-1: The Ride App shall have a maps button that presents a dropdown menu of available maps. Selecting a menu option shall open an image of the indicated map on a map's viewer page.

REQ-7-2: The route map images shall be expandable using standard "pinch-to-zoom" touchscreen interaction as well as panned for full viewing by touchscreen control.

REQ-7-3: In addition to route maps, The Ride App shall provide maps to parking around the MPEC venue, RV/camping sites at the venue, and access to the start line via bike.

## 4.8 Tradeshow venders list and booth location

### 4.8.1 Description and Priority

***The priority of this feature is 5***

There are numerous vendors of biking, running and general recreational goods which participate in a tradeshow running concurrently to the HHH riding/running events throughout the weekend. Each vendor has a booth/stall area in the MPEC building. The HHH website typically provides a map of the vendor area. The Ride App would provide users with a quickly accessible version of this map and vendors list. Furthermore, the vendors would be listed based on the category of goods. Categories would include bikes & gear, clothing, nutrition, and memorabilia.

Weekend event guests, whether riding participants or simply spectators, all have access to the tradeshow. Having a concise list of vendors by category could aid a potential customer having a better shopping experience.

### 4.8.2 Stimulus/Response Sequences

App users will access the tradeshow vendor information via a button on the app's events & info page. Selecting the "tradeshow" button will open a scrollable menu of vendor categories (bikes & gear, clothing, nutrition, memorabilia). Selection of a category will open another list of all tradeshow vendors supplying goods within that category.

### 4.8.3 Functional Requirements

REQ-8-1: The Ride App shall access a database of vendors with access fields of category of goods and vendor name.

REQ-8-2: The Ride App shall have a tradeshow menu item button which navigates to a vendor list page displaying a scrollable list of vendors based on category of goods sold.

## 4.9 List of links to local dining and lodging businesses.

### 4.9.1 Description and Priority

***The priority of this feature is 5***

Many visitors and participants in the weekend HHH activities are from outside of the immediate Wichita Falls area. Providing a listing of local restaurants and lodging choices will enhance the experience of app users and HHH participants.

### 4.9.2 Stimulus/Response Sequences

This feature will make use of a database of local restaurants, hotels, motels, and campgrounds. Associated with each business will be their address. The app will feature a page for each subcategory – restaurants and lodging. On each page, respectively, there will be a scrollable list of the businesses and the ability to click the business name. Once selected, the app will access the maps application on the mobile device, open it and direct the map app to display the business' location in google maps or apple maps as applicable.

### 4.9.3 Functional Requirements

REQ-9-1: The Ride App shall access a database of restaurants and lodging facilities with access fields of category of name and address.

REQ-9-2: The Ride App shall retrieve and display these database items in scrollable lists with hot links associated with each item. Choosing an item shall direct the app to open a dialog box to ask the user if they wish to view the choice (restaurant or hotel) in their device's map application

REQ-9-3: If desired by the user, The Ride App shall open the maps application on the mobile device and transfer the address of the chosen item to the map application.

# 5. Other Nonfunctional Requirements

## 5.1 Performance Requirements

- The system shall maintain a maximum load-time limit of 3 seconds.
- The system shall handle 1,000 concurrent users.
- The system shall maintain frame rate of at least 60 FPS for smooth UI animations.

## 5.2 Safety Requirements

- The system shall prevent data loss in the event of an app crash.
- The system shall perform regular database backups.

## 5.3 Security Requirements

- The system shall require a username and password for administrator functions.
- The system shall require first name, last name, and email address to be cross-checked with the rider database for SAG request functionality.

## 5.4 Software Quality Attributes

- The system shall be available all throughout HHH weekend.
- App shall be functional on iOS, Android, and Windows.

## 5.5 Business Rules

- A Rider must complete registration before they can receive a bib number.
- Only admin users can modify schedules and event details.
- Only admin users can assign bib numbers
- Only registered riders may request SAG assistance

# 6. Other Requirements

A minimum grade of 90% in CMPS 5153 Advanced Software Engineering for all members of the Kings of Code software development team.

# Appendix A: Glossary

HHH – Hotter'N Hell Hundred

MPEC – Multipurpose Event Center – venue for registration, tradeshow, and other HHH activities

SAG – Support and Gear – used to support riders in need of mechanical or medical help.

On-site registration – registration while at the venue shortly before participation an event.

Pre-registration – registration well prior to the start of events.

# Appendix B: Analysis Models

Preliminary case use diagram (Figure 1) reveals 3 user categories and 8 use cases for The Ride App.

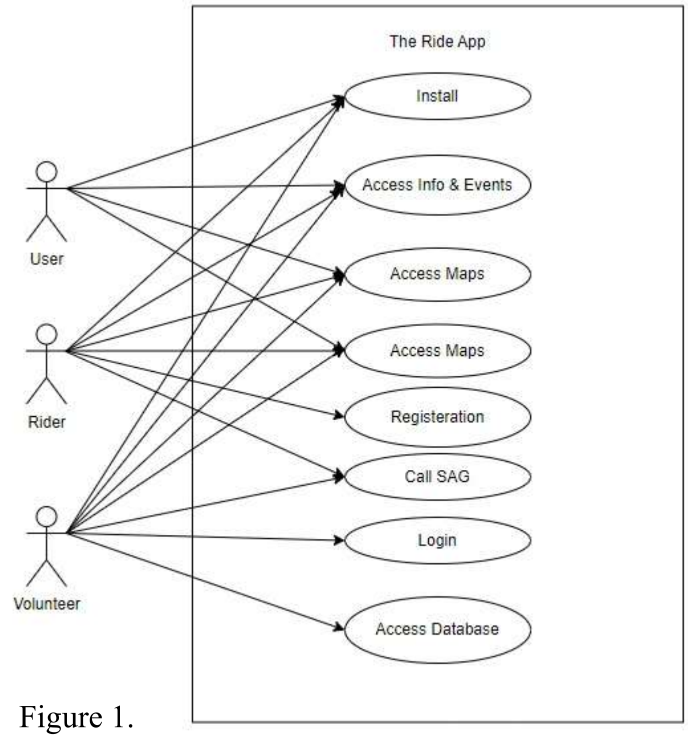


Figure 1.

Table 1. Tabular domain model for The Ride App

| Class | Attributes | Relationships | Business Rule |
|---|---|---|---|
| User | Name, Email | Registers for Event | User must have name and valid email |
| Rider extends User | Phone#, address, event, emergency contact, emergency contact phone# | Associated to Event & Registration Form | 1 Rider can register for multiple Events,<br>1 Rider can have Multiple Reg Forms<br>Can only register once for an event<br>Can access SAG feature |
| Volunteer extends User | Username, password | Access to database | Can access & modify database<br>Can access SAG feature |
| Event | Name, Date & Time, Location, | Aggregate of Registration Forms & Riders | 1 Event can have multiple Rider<br>1 Event can have multiple Reg Forms |
| Registration form | Rider, Event, Waiver, payment | Associated to Rider & Event | 1 RegForm can only be for 1 Event<br>1 RegForm can only be for 1 Rider<br>Requires a signed waiver |

# Appendix C: To Be Determined List